S1 — gain law

The truth is not a member of this one-curve family. Declared truncation, measured offline on $\bar{w}-\bar{w}_s \in [0.1,\ 25]$ from `calc_correction_functions`:

$$b_{\text{state}}{-}1: \quad 13.05\,\%\ \text{RMS}, \quad 15.34\,\%\ \text{sup}, \quad \text{range}\ [+7.79,\ +15.34]\,\%, \quad \text{one-signed}, \quad \text{far field} \to +1/8$$

The far-field limit is exact: $c_\perp \to 1+9/(8\bar{h})$ gives $b_{\text{state}} \to 9/8$. The supremum sits at $\bar{w}-\bar{w}_s \approx 1.19$, i.e. $\bar{h} \approx 2.19$, inside the canonical oscillation band — the run's most informative segment coincides with the worst point of the truncation.

The exponent 2 is $1+1/p$ at $p=1$; both asymptotic anchors give $p=1$, so it is anchored rather than chosen. No 8/9 factor is carried (user decision, 2026-08-12), which is why the declared truncation is 13 % and not 0.7 %.

S3 — what δa is, and what the model claims about it

δa is not postulated; it is what makes the law-based propagation exact, so its closed form follows by subtraction. Three properties follow from it directly and are used in S8:

(i) δa is proportional to the increment, hence exactly zero in a hold.

(ii) $1/b_{\text{state}}-1 \in [-0.133,\ -0.072]$ is one-signed, but δa is NOT: $\text{sign}(\delta a) = -\text{sign}(\text{increment})$, so it reverses with the direction of travel. Measured on the canonical scenario, $\sum_k \delta a = +0.0680$ over the descent, -0.0401 over the ascent, and EXACTLY ZERO over a closed oscillation cycle — both $\bar{a}'_{\text{true}}\mathrm{d}\bar{w}$ and $(1-\bar{a}_{\text{true}})^2\mathrm{d}\bar{w}$ are exact differentials in $\bar{w}$. The whole disturbance is acquired on the descent; a closed oscillation offers a constant-δa model nothing to lock onto.

(iii) $(1-\bar{a}_w)^2$ spans $0.83 \to 0.0019$, so δa varies over three decades.

$\delta a[k+1] = \delta a[k]$ is therefore a modelling CLAIM, not a consequence.

DECLARED, NOT DERIVED: the step from the differential law of S1 to the difference equation of S4 is forward Euler and is not stated anywhere in the main file. In the height writing that step is an error on top of a correct model; here $\bar{a}'$ reads the state, so the discretisation IS the model and the estimator is exactly estimating a slightly different family. The flow is autonomous, so an exact one-step map exists in closed form; adopting it would remove the quadrature error identically. Deferred by user decision.

S6 — the disturbance column, and a path the main file does not name

The disturbance column of F_e is the constant 1: unlike a multiplicative parameter it does not vanish at a turning point, in a hold, or in the far field. $F_e(3,5)=0$ because the control law is untouched by δa. Separability: a constant δa drives $e_{\bar{a}_w}$ as a ramp whereas an error in the gain state is a persistent offset.

y_1 ALSO CORRECTS THE GAIN. This is implied by $K = PH^\top S^{-1}$ with $H_1 = \mathbf{e}_1^\top$, giving $K_1(4) = P(4,1)/S_1$, and $P(4,1)$ is built by $F_e(3,4) = -F_{dw}$ and by Q_{34}. Nothing in the main file is wrong, but the naming ("position measurement" / "gain readout") has misled every reader including the author. Measured on the canonical scenario: $K_1(4)$ and $|P(4,1)|$ each rise by a factor ≈ 160 at motion onset (medians $1.93 \times 10^{-3} \to 3.12 \times 10^{-1}$ and $1.00 \times 10^{-7} \to 1.57 \times 10^{-5}$), while $K_2(4) \approx 7 \times 10^{-4}$ — three orders of magnitude smaller. A four-arm nested ablation over 8 seeds: removing y_2 does not reduce the across-seed spread; removing y_1's gain correction drops it 34×; removing the MA(2) memory changes almost nothing. The spread is motion-triggered, not time-accumulated (a 4× longer initial hold adds 11 % to the pre-motion spread).

That path is CORRECTLY PRICED: $K_1(4)$ is within 0.04–16 % of the realised optimum $E[e\,\nu]/E[\nu^2]$. The spread is therefore the information cost of inferring gain from position error, not a defect; removing the path costs 13.6 points of hold level.

S8 — what the constant claim omits

Q^*_{55} is computable from the S3 closed form, so what the claim omits is priced rather than assumed. Because $\delta a \propto$ increment, Q^*_{55} is zero in a hold and grows with commanded speed: a time-invariant container would be the wrong category even at the right magnitude.

FIRST ARM: $Q_{55} = 0$. That is the claim "no stochastic driving of δa", not the claim that δa is constant, which S3(ii) already refutes. Pre-registered consequence: P_{55} decreases monotonically and $\delta\hat{a}$ freezes late in the run; and since the true δa vanishes in a hold while the model keeps injecting $\delta\hat{a}$, the final hold should show a spurious ramp of order the descent accumulation.

OPEN ITEM: δa inherits the frequency content of the increment, so on a sinusoidal trajectory it is dominated by the commanded frequency. The internal model principle then says a constant model cannot reject it asymptotically; the derivable alternative is a two-state oscillator at the known commanded frequency. Not implemented.

Seed and prior

$\hat{\bar{w}}_0$ is the nominal wall and enters at the seed and nowhere else, because setting $\hat{\bar{a}}_w[0]$ IS setting the integration constant. $P_{45}[0] = 0$ because $\bar{a}(\bar{w})$ carries no δa in this writing, so $\partial\bar{a}[0]/\partial\,\delta a = 0$ and one term is complete. (The sibling `formC_state_da.tex` has a two-term $P_{44}[0]$ because there δa enters the LAW.) $P_{\delta a}$ is the only undetermined number; its contract is invariance under a 10× sweep.

floor_a as implemented is the supremum over the planned envelope (0.0306, attained at $\bar{w} \approx 2.41$), applied at a seed height where the local shape error is 0.0056. The main file does not specify which of the two is meant. Measured $\rho = \text{RMS}(e_{\bar{a}})/\sqrt{P_{44}}$ over 8 seeds is 0.15–0.18 before motion and 1.48 at the end of the run.

Results that decided each arm

Canonical scenario, 8 seeds [7 11 23 42 101 777 27 31], paired (identical draws), overall relative gain-error RMS in percent:

	desc peak	osc RMS	overall RMS
production tier-1 ($\bar{w}_s$ pinned at truth)	1.83 ± 0.72	0.96 ± 0.62	0.99 ± 0.61
production tier-2 ($\bar{w}_s$ free)	4.22 ± 1.80	1.90 ± 1.00	2.07 ± 1.38
(a) no δa	15.36 ± 6.24	4.26 ± 1.76	4.68 ± 2.56
(b) with δa	17.52 ± 6.41	7.39 ± 7.32	7.69 ± 4.89
(a) with $\bar{a}'$ from the true height	3.23 ± 0.98	1.58 ± 1.01	1.87 ± 0.93

The wall-position oracle is worth 2.1× on its own (tier-1 versus tier-2), so any comparison against tier-1 overstates the gap.

The true-slope arm improves 8/8 and is statistically indistinguishable from production tier-2 ($t = -0.48$), and better on the descent peak. Hence the derivation, the implementation and the idea of integrating the gain are all sound: the entire gap is that the integrand carries the state's own error, amplified by $\partial\bar{a}'/\partial\bar{a} = -2(1-\bar{a})$. Production also integrates — its predict is $x_4 + \bar{a}'(\cdot)$ exactly as here — so it does not re-anchor; the only structural difference is where $\bar{a}'$ is evaluated.

The additive constant disturbance is falsified in four configurations: harmful on canonical with either slope source, harmful on the Menq ramp ($\bar{h}\,6.67 \to 1.111$ over 10 s, near-wall gate opened), and $\delta\hat{a}$ never leaves noise ($-5.4 \times 10^{-8} \pm 5.3 \times 10^{-6}$ against a prior of 1.53×10^{-4}). Under the true slope there is no model error left to absorb, so δa can only fit noise and its harm there is expected rather than evidence about the concept.

Breaking the feedback by lagging the slope does not work: evaluating $\bar{a}'$ at the pre-update state changes nothing ($+0.076$, $t = 0.65$) and smoothing is monotonically harmful. So the gap is the BIAS in $\hat{\bar{a}}_w$, not the noise on it.

Menq's forgetting factor (thesis eq. 4.15, $P = \lambda_f^{-1}[I - KH]P^f$) is falsified on 12/12 paired comparisons across two trajectories, two arms and $\lambda_f \in \{0.9999, 0.999, 0.99\}$. It does repair the late over-confidence (ρ 1.29 → 0.79) but pushes the early over-pessimism further down: $\rho(t)$ here is not uniformly biased, and a uniform-in-time inflation cannot help a deficit whose sign changes with time.